AI/ML PROJECT SUBMISSION

PROJECT OVERVIEW	STUDENT DETAILS
• Project Number: 3 • Task Name: Expense Tracker System. • Difficulty: Moderate	• Name: Pranav Gajanan Rane • College: Government Polytechnic Gondia • Branch: Computer Engineering • Domain: AI/ML

```
from datetime import datetime

def add_expense(): #add expenses made
 try:
  description = input("Describe the expense: ")
  category = input("Category of the expense: ")
  date = input("Date of expense (DD-MM-YYYY): ")
  # validate date format early
  try:
   datetime.strptime(date, "%d-%m-%Y")
  except ValueError:
   print("Date must be in DD-MM-YYYY format.")
   return

   amount = float(input("Enter the amount you spent: "))
  if amount <= 0:
          print("You cannot make negative or zero expenses!")
   return

  expenses.append({
   "desc": description,
   "category": category,
   "amount": amount,
   "date": date
  })
  print("Expense added successfully!")

 except ValueError:
```

```
 print("The book is available to be issued!")
library[issue]['available'] = False
      name = input("Enter your name - ")
      library_id = int(input("Enter your library id - "))
      library[issue]['borrower_name'] = name
      library[issue]['borrower_id'] = library_id
      today = date.today()
      library[issue]['issue_date'] = today
      due_date = today + timedelta(days=7)
            print(f"You need to return the book by -
{due_date}")
          print(f"You, {library[issue]['borrower_name']}
have successfully borrowed the book!")
      library[issue]["due_date"] = due_date

     else :
      print("The book is unavailable to be borrowed! ")

   else :
     print("The book with this ISBN isn't available in our
library!")
  except ValueError :
   print("Pls enter correct value!")


def return_book() : #Returns a book
  try :
    issue = input("Enter the ISBN of book to be returned
- ")
```

```
except ValueError:
    print("Please enter correct values (amount must be a number).")


def view_expenses(): # shows all expenses made
 if not expenses:
  print("No expenses made!")
 else:
  for index, expense in enumerate(expenses, start=1):
   print(f"{index}) {expense['desc']}")
   print("Expenses made for this -")
   print(f"Category - {expense['category']}")
   print(f"Amount - Rs.{expense['amount']:.2f}")
   print(f"Date - {expense['date']}")
   print("=" * 30)


def category_summary(): # showcases category-wise spending
 if not expenses:
  print("No expenses made!")
  return

 c_summary = {}
 for expense in expenses:
  cat = expense["category"]
      c_summary[cat] = c_summary.get(cat, 0.0) + expense["amount"]

 print("Category-wise summary:")
 for category, total in c_summary.items():
```

```
print(f"{category}: Rs. {total:.2f}")

def budget_report(budget): # Cslcukate budget report
    total_spent = sum(exp["amount"] for exp in expenses)
 print(f"Total spent: Rs.{total_spent:.2f}")
 print(f"Budget: Rs.{budget:.2f}")
 if total_spent > budget:
      print("WARNING!! - You have exceeded your budget!")
 else:
  remaining = budget - total_spent
   used_pct = (total_spent / budget) * 100 if budget > 0 else 0
      print(f"You are within budget. Remaining: Rs. {remaining:.2f}")
  if used_pct >= 100:
            print("You have used 100% of your budget(overbudget).")
  elif used_pct >= 80:
     print(f"WARNING: You have used {used_pct:.2f}% of your budget (>=80%).")
  else:
    print(f"Budget used: {used_pct:.2f}%.")


def top_category(): # most expensive category
 if not expenses:
  print("No expenses made yet.")
  return
```

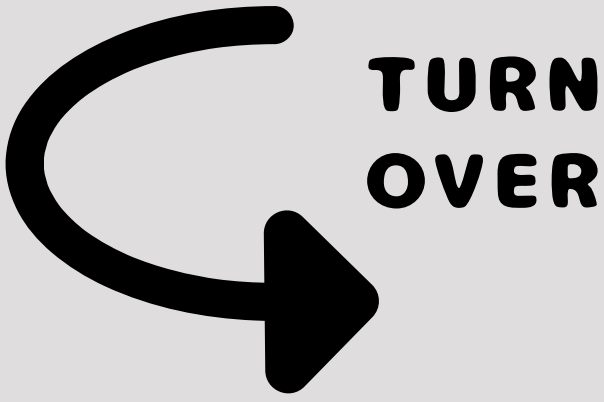
```
 summary = {}
  for exp in expenses:
    category = exp["category"]
     summary[category] = summary.get(category, 0.0) +
exp["amount"]

  top = max(summary, key=summary.get)
  print(f"Top Category: {top} (Rs.{summary[top]:.2f})")



def most_expensive(): #calculate the most expensive
purchase made
  if not expenses:
    print("No expenses recorded yet.")
  else:
          max_exp = max(expenses, key=lambda x:
x["amount"])
    print(f"Most expensive expense: {max_exp['desc']} |
Rs.{max_exp['amount']:.2f} | {max_exp['date']}")



def average_daily_spending(): #bonus task
  if not expenses:
    print("No expenses made!.")
    return

  dates = []
  for exp in expenses:
    try:
          d = datetime.strptime(exp["date"], "%d-%m-
%Y").date()
```

```
dates.append(d)
    except ValueError:
      print("Enter correct date!")
      continue

  if not dates:
    print("No valid dates found in expenses.")
    return

  min_date = min(dates)
  max_date = max(dates)
  days_span = (max_date - min_date).days + 1
      total_spent = sum(exp["amount"] for exp in
expenses)

  avg_per_day = total_spent / days_span if days_span
> 0 else total_spent
   print(f"Expense date range: {min_date.strftime('%d-
%m-%Y')}      to      {max_date.strftime('%d-%m-%Y')}
({days_span} day(s))")
  print(f"Total spent: Rs.{total_spent:.2f}")
    print(f"Average daily spending over this range: Rs.
{avg_per_day:.2f}")


def show_menu():
  print("\nWhat you can perform - ")
  print("1) Add Expense")
  print("2) View all expenses")
     print("3) See total expense for each category
(summary)")
  print("4) Calculate your Budget Report")
```

```
print("5) Find the most expensive single expense")
  print("6) Show top spending category")
  print("7) Calculate average daily spending")
  print("8) Exit")


'''Actual System Below '''

expenses = [] # globally stores


def main():
  try:
    budget = float(input("Set your monthly budget: "))
    if budget <= 0:
      print("Budget must be positive!")
      return

    while True:
                    print("\n_____Welcome    to    Budget
Tracker_____")
      show_menu()
      choice = input("Enter what you want to do (1-8) - ")

      if choice == "1":
        add_expense()
      elif choice == "2":
        view_expenses()
      elif choice == "3":
        category_summary()
      elif choice == "4":
```

```
budget_report(budget)
      elif choice == "5":
        most_expensive()
      elif choice == "6":
        top_category()
      elif choice == "7":
        average_daily_spending()
      elif choice == "8":
        print("Exiting...")
        break
      else:
        print("Enter valid choice only (1-8)!")
  except ValueError:
    print("Please enter correct values for budget.")


if __name__ == "__main__":
  main()
```

Note - This code might break due to incorrect indentation while pdf making so pls use the code mentioned in .py file attached



TEST OUTPUTS -

```
_____Welcome to Budget Tracker_____

What you can perform -
1) Add Expense
2) View all expenses
3) See total expense for each category (summary)
4) Calculate your Budget Report
5) Find the most expensive single expense
6) Show top spending category
7) Calculate average daily spending
8) Exit
Enter what you want to do (1-8) - 1
Describe the expense: For health
Category of the expense: Health
Date of expense (DD-MM-YYYY): 23-2-2026
Enter the amount you spent: 400
Expense added successfully!
```

```
Enter what you want to do (1-8) - 2
1) For health
Expenses made for this -
Category - Health
Amount - Rs.400.00
Date - 23-2-2026
==============================
2) Food
Expenses made for this -
Category - Food
Amount - Rs.100.00
Date - 24-2-2026
==============================
3) Health
Expenses made for this -
Category - haelth
Amount - Rs.7341.00
Date - 31-3-2026
==============================
```

```
_____Welcome to Budget Tracker_____

What you can perform -
1) Add Expense
2) View all expenses
3) See total expense for each category (summary)
4) Calculate your Budget Report
5) Find the most expensive single expense
6) Show top spending category
7) Calculate average daily spending
8) Exit
Enter what you want to do (1-8) - 3
Category-wise summary:
Health: Rs. 400.00
Food: Rs. 100.00
haelth: Rs. 7341.00
```

```
_____Welcome to Budget Tracker_____

What you can perform -
1) Add Expense
2) View all expenses
3) See total expense for each category (summary)
4) Calculate your Budget Report
5) Find the most expensive single expense
6) Show top spending category
7) Calculate average daily spending
8) Exit
Enter what you want to do (1-8) - 4
Total spent: Rs.7841.00
Budget: Rs.10000.00
You are within budget. Remaining: Rs.2159.00
Budget used: 78.41%.
```

```
Enter what you want to do (1-8) - 4
Total spent: Rs.9541.00
Budget: Rs.10000.00
You are within budget. Remaining: Rs.459.00
WARNING: You have used 95.41% of your budget (>=80%).
```

```
Enter what you want to do (1-8) - 5
Most expensive expense: Health | Rs.7341.00 | 31-3-2026
```

```
Expense date range: 23-02-2026 to 31-03-2026 (37 day(s))
Total spent: Rs.9541.00
Average daily spending over this range: Rs.257.86
```

TEST OUTPUTS IN BROKEN CASES -

```
Date of expense (DD-MM-YYYY): 8
Date must be in DD-MM-YYYY format.

_____Welcome to Budget Tracker_____
```

CODE TESTING WAS DONE SUCCESSFULLY!

More Details about the project is mentioned in the .md file attached . This pdf was made specifically to attach the code and its outputs.

PROJECT COMPLETED